Roadmap for C# v14

Note: Each item contains a link, which can be accessed by opening the document in a web browser.



Learn the Fundamentals

- Basic Syntax
- **Variables and Data Types**
  - String
  - Int
  - Boolean
  - Float/Double/Decimal
  - Dates and Times
  - Guid
  - Var
- Conditionals
- Methods
- Enums
- **Collections**
  - List
  - Dictionary
  - Array
  - Collection Expressions
- Loops
- Namespaces
- `Using` directives
- String Interpolation

Object-Oriented Programming

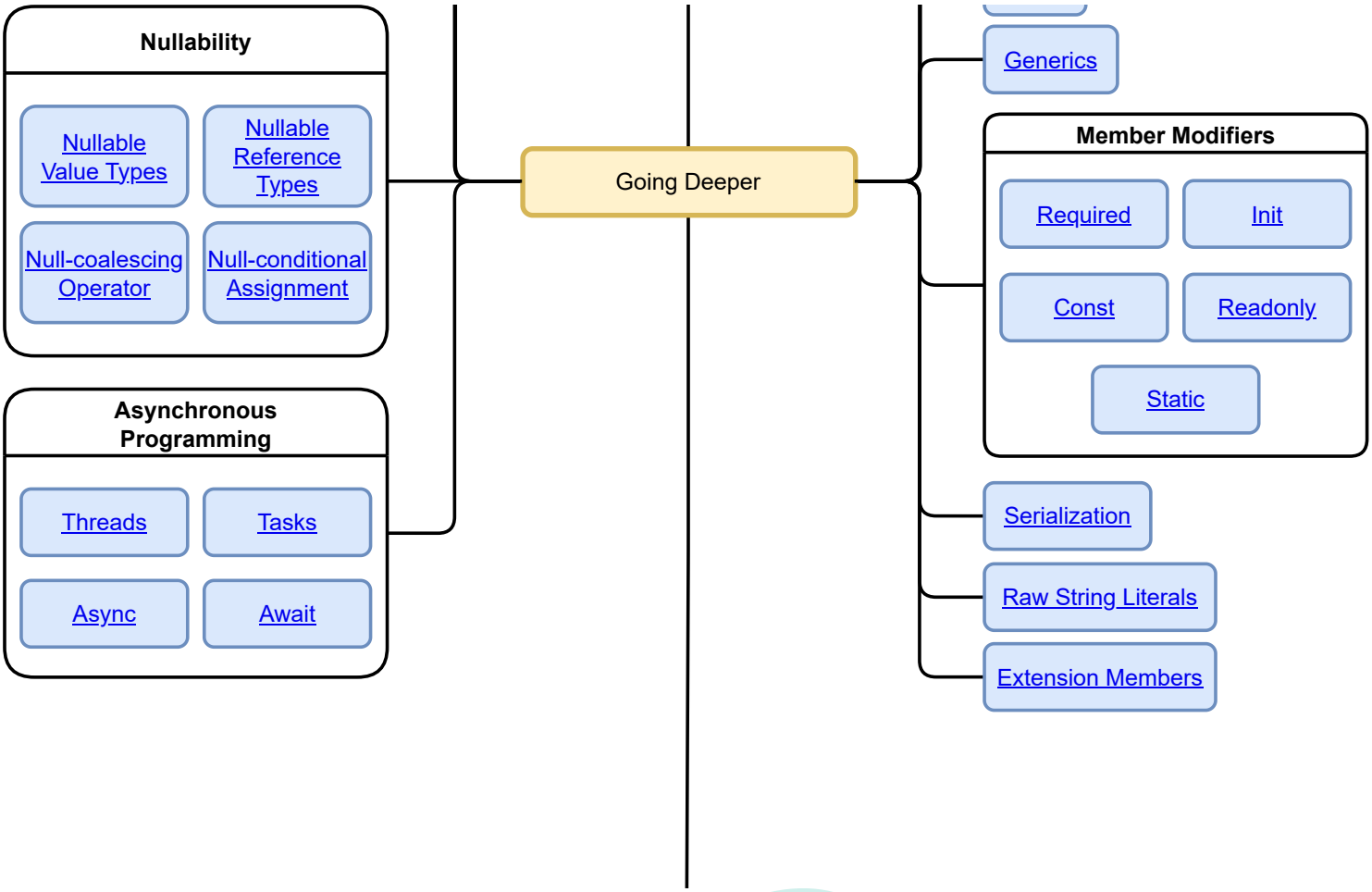
- **Encapsulation**
  - Classes
  - Structs
  - Records
  - Primary Constructors
- **Access Modifiers**
  - Public
  - Private
  - Internal
  - Protected
- **Inheritance**
  - Interfaces
  - Abstract Classes
- **Polymorphism**
  - Virtual
  - Override
  - Using Base Classes

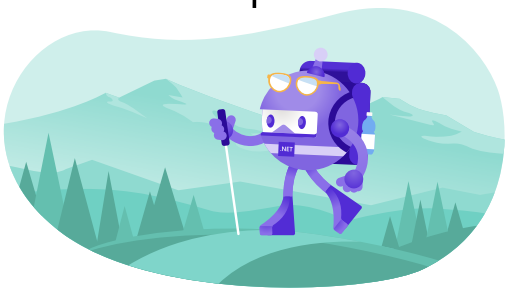
Exception Handling

- Exception Handling
- Object and Collection Initializers

LINQ

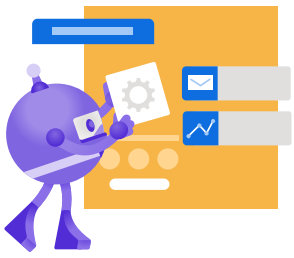
- **Lambdas**
  - "=>" Operator
  - Arrow Functions
- LINQ





Moving on to .NET

Congrats! Now that you have a solid understanding of the language features of C#, the .NET platform has many options for your next path.



Web Development

Create websites, web applications, and back-end APIs with ASP.NET

https://dotnet.microsoft.com/en-us/apps/aspnet



Native App Development

Create installable desktop and mobile applications with Avalonia UI or .NET Multi-Platform App UI (.NET MAUI)

Avalonia UI: https://avaloniaui.net/



.NET MAUI: https://dotnet.microsoft.com/en-us/apps/maui



Game Development

Create video games with Unity, Godot, MonoGame, or Stride

Unity: https://dotnet.microsoft.com/en-us/apps/games/unity

Godot: https://godotengine.org

MonoGame: https://www.monogame.net/

Stride: https://www.stride3d.net



Machine Learning

Create data-driven models to guide decision-making with ML.NET

https://dotnet.microsoft.com/en-us/apps/machinelearning-ai/ml-dotnet